

Ms. Ananya Roy

Located in Toronto, Canadian Citizen, Open to Relocation, References on Request

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Summary:

- Experienced technologist, passionate Generative AI practitioner and ML/data science leader with 15 years of broad industry experience in machine learning, delivering high-impact AI/ML solutions driving profitability across financial services, sports betting, and e-commerce
- Proven expertise in building, deploying sports trading models, dynamic pricing models, revenue optimization models, personalization, with a strong foundation in advanced statistical modelling, bayesian methods, machine learning as well as new technologies like Generative AI, agentic frameworks with a constant learning mindset
- Inventor on multiple filed and granted patents in ML and Generative AI, spanning agentic systems and probabilistic modelling
- Strong track record of providing technical leadership in solving complex problems, understanding business objectives and aligning with ML development and deployment of scalable ML systems
- Consistently produced results on high value projects from conception to implementation, production that stood the test of time, in various capacities, contributing individually and leading, mentoring cross-functional teams
- MSc in Applied Math, London School of Economics (LSE), MSc,BSc in Pure Math, Delhi University

Skills:

Machine Learning - Generative Text AI (LLM's, RAG, fine-tuning, vector databases, Agentic Frameworks - langgraph, crewai, libraries like hugging face, langchain, etc), MCP, Traditional Machine Learning Algorithms, Deep Learning (Transformers, RNN, etc.), NLP, Bayesian/Probabilistic Models (in STAN, MCMC methods);

Languages and Frameworks - Python, R, STAN, C++, MATLAB/Octave; Frameworks – TensorFlow, Keras, PyTorch, Scikit-Learn, PySpark; Databases - SQL, Snowflake, MongoDB, pgVector, chromaDB; AWS/Google Cloud Platform/Openshift; REST API's development, End to end deployment, CI/CD, Docker, Airflow

Leadership Skills - Team building & hiring, roadmap and AI strategy, cross-functional delivery, stakeholder management, effective communication(SVP/CTO), mentoring & career development

PROFESSIONAL EXPERIENCE

Director and Principal Engineer, Compliance AI Solutions, Royal Bank of Canada **Nov 2025 – present**

Member of Technical Distinction Program - an elite coterie of select technologists with direct access to CTO to define tech strategy

- **Stood Up the Compliance AI/ML function from scratch in Nov 2025** - defined the operating model, hired and leading a team of 5 - senior MLE's and software engineers, and built a delivery roadmap that has moved four programs from concept to production in under a year
- Hands-On technical leader supporting the team in technical development, code reviews, brainstorming novel solutions and mentoring junior talent to support their career development
- Shipped an agentic AI (langgraph) system that autonomously triages regulatory changes, determines applicability to RBC, writes triage decisions back through vendor's API's **eliminating \$2M+ in external consulting fees**
- Delivered a news aggregation, categorization and summarization tool with MNPI detection for trade surveillance analysts(~30 users), integrated with Kx, **cutting alert resolution times from 45mins to under 20mins**
- Leading development of an intelligent Email Management and Workflow Automation tool for the Registrations team, classifying inbound emails and triggering automated workflows – **replacing manual processing by a team of 4-5 FTEs**
- Completed a POC for multilingual e-communications translation and insider trading signal detection which **outperformed the incumbent vendor developed solution**. Approved for production next year and set to replace the existing vendor solution leading to significant savings
- **Own the full delivery lifecycle for every project** from problem framing and prototyping, architecture and model selection, CI/CD, and enterprise model risk review to production deployment

- **Drive stakeholder relationships across technology and Business**, translating compliance pain points into a funded, prioritized AI roadmap

Director, AI Solutions, RBC Borealis, Royal Bank of Canada

Sep 2024 – Nov 2025

- **Delivered a Meeting Prep Agentic AI solution for Wealth Management advisors**, generating automated pre-meeting reports surfacing client investment portfolios, KYC data, and trade history; drove end-to-end production deployment in close collaboration with Product and Business teams, incorporating iterative feedback from financial advisors to refine outputs and ensure adoption
- Supported the **production launch of a Meeting Transcription Agent**, navigating the full enterprise Model Risk Management (MRM) approval process to meet RBC's rigorous governance and validation standards; both solutions contribute to RBC's AI & Digital Transformation program targeting **\$100M in business impact**

Director, Data Science & Principal Technologist, Royal Bank of Canada

April 2021 – Sep 2024

- **Led several Generative AI initiatives to optimize and automate business processes**, finding opportunities and business partners and executing - from ideation, development to production; managing business relationship and securing executive buy-in resulting in CAD 18million in savings in process transformation
- Spearheading work on several **patents in Generative AI**, **presenting work at townhall to over 1000 people**
- Demonstrated **strong stakeholder management** by consistently engaging with diverse business and technology counterparts across RBC, including Wealth Management, Engineering, and Product teams
- Maintained **regular and transparent communication with stakeholders** through structured updates, demos, and status check-ins, ensuring alignment on priorities and timely delivery of complex GenAI initiatives
- Built trusted relationships with business partners by effectively translating technical concepts (e.g., LLMs, agentic frameworks, probabilistic models) into business value, which helped secure executive sponsorship and business buy-in for multiple projects
- Concept to implementation / production of AI/ML products in collaboration with Engineering, Product teams
- Developed and deployed internal AI/ML solutions, like building a semantic layer on top of all data sources using only metadata, used extensively by products across the bank and one of the first to onboard to Cloud
- Built a reusable **Marketing Mix Modelling (MMM) prototype for optimizing marketing/advertising spend** budgets, leading to cost savings of > CAD 1 million annually; Automated the dispute management process with an ML solution, significantly reducing processing time and the number of FTE agents handling disputes
- Regular technical presenter at internal and external conferences on LLM's and Prompt engineering, bayesian models, etc, featured in Inventor's Spotlight for patents, initiated research paper reading groups

Senior Manager, Data Science, Royal Bank of Canada

May 2018 – Mar 2021

- **Automated Credit Adjudication process to reduce decision time from 40 minutes to under 2 minutes** for credit applications in prediction in a decision in real-time, leading to significant cost savings and improved efficiency; patent filed for our proposed solution
- **Predicting risk ratings for funds** (ETFs, Mutual Funds, etc.) using publicly available and internal RBC documents, saving 60% of human hours spent by risk analysts; worked on Text classification, summarization tool to summarise legal, financial documents for Group Risk Management
- Automated processes for regulatory requirements and facilitating decision-making by applying Named Entity Recognition to identify economic interdependencies, relationships between clients from public sources (web scraping) and internal RBC documents; patent filed
- Developed enterprise-wide educational content on ML for better AI adoption and understanding

Patents

- Full Patent - [System & Method for Machine Learning Architecture for Interdependence Detection](#)
- Other patents in provisional stage - Gen AI Planning and Executions Agents, proposed a probabilistic hierarchical bayesian approach to model credit adjudication, etc

Machine Learning Algorithm Developer, Pinnacle.com

July 2016 – May 2018

R&D /Quant arm of Pinnacle Sports with best-in-industry betting prices and one of the most sophisticated sports-betting models

- Developed and deployed **dynamic sports betting pricing models in production**, directly optimizing betting odds and maximizing profitability while minimizing cannibalization across betting markets, with **typical flux in**

billions of dollars, profitability > \$10 million with great feedback from human traders on efficacy of models in production

- Built and deployed a **hierarchical Bayesian Customer Sharpness model used in pricing betting odds** which outperformed previous models in production by ~50%
- **Scaled and automated production pipelines from 2 sports to 20 within a year** – extracts and clean data from SQL databases, runs bayesian models, generate training and evaluation reports weekly for top 20 sports using AWS
- Built scalable, automated pricing pipelines and **monitoring customer behaviour and market elasticity in near-real-time**, enabling rapid model updates and agile response to changing demand conditions
- Developed and productionized **soccer live injury models** with proven impact - savings in millions of dollars
- Worked on Soccer Live Arbitrage Model for the discovery of arbitrage opportunities and updating betting markets
- Used a range of frequentist ML approaches like Random Forests, Decision Trees, Neural Networks and Bayesian/probabilistic models on a range of problems; automated reporting framework for some big sports using AWS
- Conducted Trainings/ Talks for members of the Research and Development Team and a wider audience

Data Scientist, Cardekho.com

April 2015 – April 2016

- Led the development and deployment of the **Used Car Pricing Engine - a dynamic pricing model** integrating market trends, vehicle attributes, and user demand signals, serving as the core pricing backbone for multiple products including insurance and other offerings
- The Used Car Pricing Engine consistently got a **high 4-star user rating and 80%+ user satisfaction**
- Developed a scalable and dynamic **personalized search and recommendation system** leveraging user search and lead behaviour historical data, significantly increasing lead generation and customer engagement through individualized price and product displays and a personalized user browsing experience
- Reporting directly to CEO, played a key role in expansion of the Data Science team as a leader/mentor

Research Consultant, Thought Arbitrage Research Institute

Mar 2013 – Aug 2014

- Developed **land pricing model for Government of India** in collaboration with German think tank GIZ, with GoI expressing interest in undertaking similar projects on a bigger scale for possible implementation of this pricing model in rural India
- Secured several additional projects from large private clients on land pricing
- Data extraction, cleaning and analysis of Consumer Expenditure surveys by NSSO and gross all-India production survey data by ASI; conducted impact assessments (SROI) of CSR projects

ACADEMICS

MSc, Applied Math, London School of Economics, UK

2011 – 2012

- Recipient of Advent Software Scholarship for 2011-12 for an amount of GBP 10,000
- Top 5 for Bernard Levin Award for student journalism, 2012 sponsored by BBC, LSE Alumni and Students' Union
- Key Modules – Game theory, Stochastic Analysis, Financial Risk Analysis and Quantitative Methods in Finance, Algorithms

Research Scholar, Applied Statistics Unit, Indian Statistical Institute

2010-2011

- Research on Integral Value Transformations (IVT) and its applications in cryptography, evolution of DNA sequences, etc
- Published two papers as co-author

MSc, Pure Mathematics, Delhi University

2008 – 2010

- Key modules: Topology, Measure Theory, Complex Analysis, Functional Analysis, Calculus on R^n , etc

BSc (Hons.), Mathematics, Delhi University

2005 – 2008

- Key modules: Analysis, Algebra, Differential Equations, Probability and Mathematical Statistics, etc